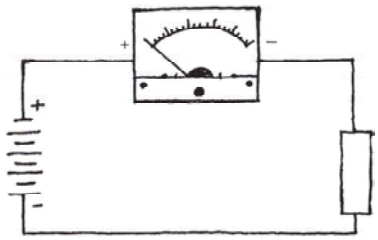
Chapter 6

Electric Current and Circuit



In your previous classes, you had learned to make some simple electrical circuits. Can you make an electrical circuit using only a wire, a single bulb and a battery? In how many different ways can you connect the above components? Remember that you have to use only a single piece of wire without cutting it.

In this chapter, we shall study several concepts of electricity. You already know that electricity exists in two forms, static electricity and electric current. Static electricity is that form of electricity where the electric charge remains stationary. While in an electric current, the electric energy flows due to the potential difference between two points in the circuit. We have studied static electricity in previous classes. We shall now learn about current electricity in this chapter.

6.1 Electric Current

Let us try to understand the concept of electric current through an activity:

Activity-1

Required materials – connecting wires, iron nail, wooden twig or a pencil, electric bulb and a cell.

Case 1: Form a circuit, as shown in figure-1(a), using connecting wires. Connect the bulb, nail and cell together. After completing the circuit check the state of the bulb. Does the bulb light up?

Case 2: Now, replace the iron nail with a wooden pencil in the circuit, as shown in figure-1(b). In this case, does the bulb light up on completion of the circuit? If not, why?

Case 3: In the circuit, leave a gap between the connecting wires, as shown in figure-1(c).What happens to the bulb, does it light up?

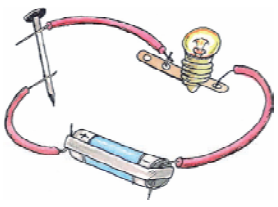


Figure-1 (a)

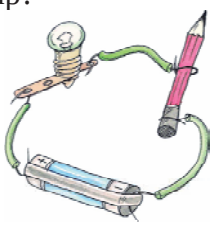


Figure-1 (b)

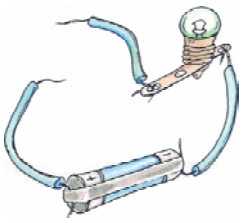


Figure-1 (c)

You can also use a coin, a glass rod, paper, rubber, or anything else in place of the nail and repeat the above observations.

We observed that, when the circuit is closed using connecting wires, or if any conducting material is present with the connecting wires in the circuit, then the bulb lights up. In this case, the electric current flows through the circuit without any obstruction. But if the circuit is broken, that is if the connecting wires are not connected properly or a non-conducting material is present, then the current does not flow through the circuit and hence the bulb does not light up. Group the materials used by you in the above experiment into two categories, conductors and non-conductors.

We can state that conductors allow the flow of electric current through them, while, non-conductors or insulators prevent or resist the flow of current. Remember, the flow is not of electric current but of electric charges, which in turn produces electric current. However, for simplicity, we call it flow of current. This is similar to the water current produced due to the flow of water molecules. Here, the quantity of water molecules is analogous to the quantity of electric charge. We shall try to understand some concepts of electricity using similar analogy.

Remember that unlike the molecules of water, molecules of current do not exist. This is because electric current is not matter but is a form of energy. Therefore, though you can see the flow of water through a cut pipe, you cannot see the flow of electricity through a cut wire, rather cutting a wire stops the flow of current.

Electric current is the total electric charge that flows across a cross section area of a conductor, in a unit of time. The measuring unit of electric charge is Coulomb. If 'Q' Coulomb charge flows in 1 second across a conductor, then the quantity of electric current flowing across the conductor would be,

Current, $I = Q/t$ …….. (1)

The SI unit of measurement of electric current is Ampere, represented as 'A'. This is a vector unit.

From equation (1), we can derive that when 1 coulomb of charge flows across a cross section of conductor in 1 second, the current flow is defined as 1 ampere.

1 ampere = 1 coulomb/1 second



French Scientist Sir Andre-Marie Ampere (1775-1836) is known as the 'father of electrodynamics '. The unit Ampere is named after him.

Sometimes, smaller units than ampere are used like milliampere (mA) or microampere (μA).

1 mA = 1/1000 A = 10^{-3} A

1 μA = 1/1000000 A = 10^{-6} A

Electric current in an electrical circuit is measured using an Ammeter. It is an instrument which shows the quantity of electric current flowing.The positive terminal of the ammeter is attached to the Positive terminal of the cell and the negative terminal to the negative in a series combination. Ammeter has very low resistance to current and hence is attached in series combination in the circuit, so that maximum current can flow through it.

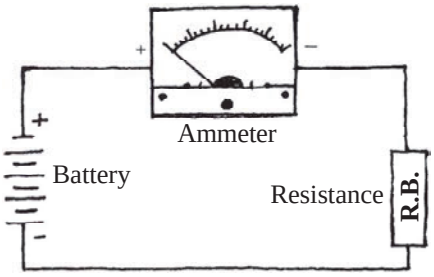


Figure- 2: Ammeter connected

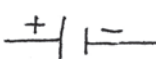
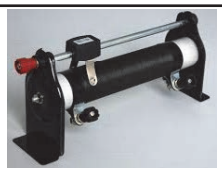
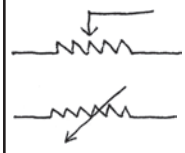
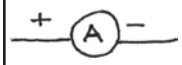
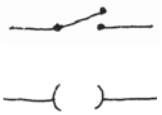
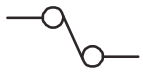
Questions:

1. If a 0.4A electric current flows in a circuit for 10 minutes, then, calculate the total charge flowing in the circuit.
2. Write the definition for the unit of measurement of electric current.

6.2 Components of an Electrical Circuit

Electrical components like connecting wires, electric bulbs, switch, ammeter, voltmeter etc.,can be parts of an electrical circuit. We usually show an electric circuit by drawing a line diagram. The various components of a circuit are shown by the signs, as listed in (table-1). Try to collect these components and any other that you can find, and study and tabulate them.

Table-1: Signs used in circuit diagrams

Electrical component	Picture	Sign	Electric component	Picture	Sign
cell			Battery		
Connecting wires			Resistor		
Joining wires			Variable resistor		
Electric bulb			ammeter		
Key/switch on			voltmeter		
Key/switch off			fuse		

Questions:

1. Can you tell whether a current is flowing in an electric wire by observing it? If not, how will you test the flow of current?
2. Why should we use items made of wood or rubber to detach a live wire from an electrocuted person?

6.3 Electric Potential and Potential difference

We have observed in our day to day life that water always flows from a container having a higher level to the container having a lower level, till the water level of both become equal. Normally, this water flow is due to the pressure difference, and it continues till the pressure in both the container becomes the same. In the above case, the water flows automatically, from higher to lower pressure as soon as the two containers are connected.

Similarly, for an electric current to flow in any circuit, the presence of potential difference is necessary. Due to difference in electric potential in a circuit, an electric charge flows from the point at a higher potential towards the point at a lower potential. This flow of charge is known as electric current. To maintain the flow of electric current, we have to create a potential difference in the circuit by using a source of voltage. Electric cell and battery are types of voltage source.

Electric Field

The area surrounding a charge, within which it experiences a force of attraction or repulsion due to the presence of another charge, is known as the electrical field of the charge. If a positive charge 'Q' is kept at a point, an electric field is created around it. The intensity of the field is strongest near the charge and weakens as the distance from the charge increases.

The force experienced by a single positive electric charge positioned at any point in an electric field is known as the intensity of the electric field at that point. It is represented by 'E' and its unit is Newton/Coulomb.

If a positive charge 'q', experiences a force F, then the force for a unit (singular) of charge would be F/q, thus,

Intensity of electric field, E=F/q

The electric potential at point P in an electric field is equal to the work done to bring a unit positive charge from infinity to that point P. If W is the amount of work done to bring a charge 'q' from infinity to point 'P' of the electric field, then work done to bring a unit charge from infinity to that point would be W/q.

Potential $V=W/q = \frac{\text{work done}}{\text{charge}}$ ..... (2)



The unit volt is named after the Italian physicist Alasandro Volta.

The unit of measurement of potential in SI system, Volt = Joule/ Coulomb.

Similarly, the potential difference between two points in a circuit is the amount of work done to move a unit positive charge between the two points. The potential difference is measured between two points of a circuit by an instrument, voltmeter. The positive terminal of the voltmeter is attached to the positive of the cell/battery and the negative of the voltmeter to the negative terminal of battery, in a parallel combination.

The unit of measurement of potential difference in the SI system is Volt, denoted as 'V'. We know that the electric potential in a cell or battery is higher at the positive terminal than the negative terminal. When we move from the positive to the negative terminal of the battery, in an electric circuit, there is a decline in the potential difference which is known as voltage drop.

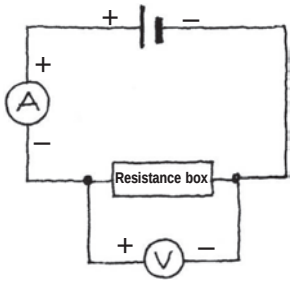


Figure-3: Voltmeter in a circuit

6.3 The Ohm's Rule



George Simon Ohm

In 1827, the physicist George Simon Ohm, established the relation between the potential difference, resistance and electric current in a circuit. We shall undertake the following activity to understand it-

Activity-2

Make an electric circuit as shown in Figure-4. The components needed are – a 0.5 meter long nichrome wire (XY), an ammeter, a voltmeter, four electric cells of 1.5 volts each.

We shall use only one cell in the beginning. To measure the current 'I', flowing across XY, an ammeter is used and a voltmeter is used to measure the potential difference 'V', between the two cell terminals. Note down the ammeter and potential difference readings in table-2.

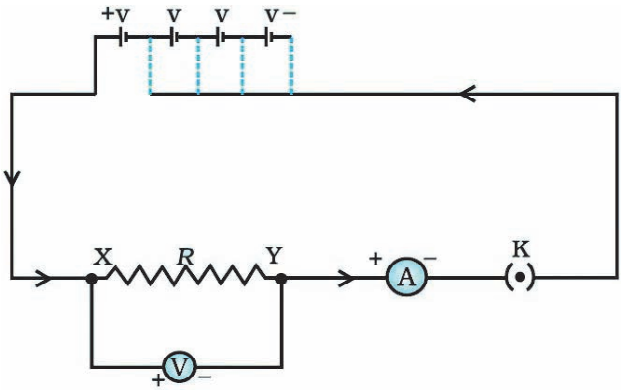


Figure-4: Circuit to demonstrate Ohm's rule

Now join two cells in the circuit and note the values of potential difference and current as reading-2. Similarly, we join 3 cells and four cells and note the measurements as readings-3 and 4, respectively, in the table.

Table-2

S.N.	No.of cells in the circuit	Current I, flowing in wire XY	Potential Difference V, between X and Y	V/I volt
1.	1			
2.	2			
3.	3			
4.	4			

- What are the values of V/I for each pair of potential difference V and current I?
- Draw a graph between the values of V and I and observe the relation between them.

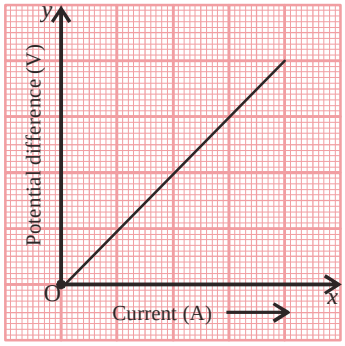
In this activity, you will find that the value of V/I for each reading remains almost constant.

This is shown in graph-1, where V/I is a straight line drawn from the initial point. It means, V/I, is a constant ratio. Therefore, V∝I.

V = RI (where R is a constant)

Or, R = V/I or I = V/R .....(3)

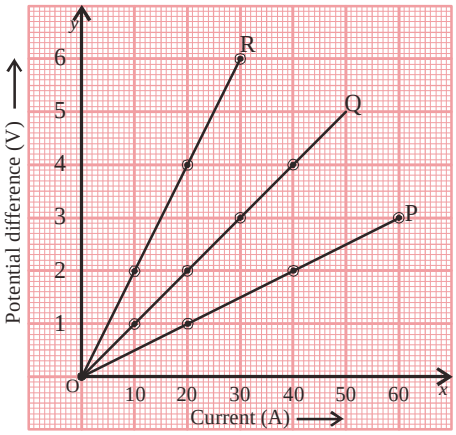
Here R is a constant which is known as the resistance of the wire. The measuring unit of resistance in the SI system is Ohm (Ω). From eq. (3), if the potential difference between two terminals of a conductor is 1 V and a current of 1 A, flows in it, then the resistance R of the conductor is said to be 1 ohm.



Graph 1

$$1 ohm = \frac{1 Volt}{1 Ampere}$$

It is clear from eq. (3), that the current flowing in a resistor is inversely proportional to the value of resistance R. Thus, if the potential difference remains constant, the amount of current falls to half when the resistance is doubled. In practice, it is often necessary to increase or reduce the current in a circuit without changing source voltage, and this is done by using a component called variable resistor. It is also known as rheostat (current controller).



Graph 2

Question :

The values of potential difference and current in three resistors P, Q and R are shown in graph-2. Find out the resistances of the resistors and establish the ratio between them, i.e., P:Q:R.

6.4 Resistance and Conductance

You must have observed that in the electric circuits in our homes, in some places thick electric wires are used and in other places thinner wires are used. You know that the amount of current flowing in a circuit depends on the potential difference in the circuit as well as the resistance offered by the circuit. In wires with equal cross section thickness, the resistance offered by the longer wire is more than that offered by the shorter wire. Similarly, in wires of equal length, the thicker wire will have a lower resistance than the thinner wire.

Therefore, current flow is easier in a circuit having thicker and shorter wires, and hence in our homes attention is given to the thickness and length of the wires used in the circuits. The resistance in a conductor depends on the following –

- length of the conductor
- cross section area of the conductor
- property and nature of the conductor material
- temperature of the conductor

Activity-3

(Part 1 – Length of the conductor)

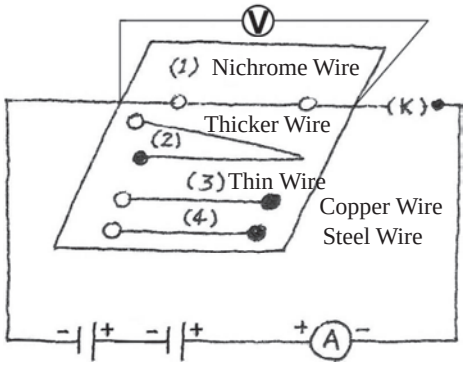


Figure- 5: Study of resistance

- Take a 100 cm long Nichrome wire.
- Mount the two ends of the wire tightly, checking that the wire is not coiled.
- Now attach two cells of 2 V each in series, a voltmeter and ammeter to complete the circuit.
- Measure the current, I and the voltage, V across the Nichrome wire and fill the values in table-3(a).
- Now using the mount, change the legth of the wire to 80, 60, 40 and 20 cm and fill table-3(a), accordingly. Draw a graph between the values of resistance and length of wire and establish the relation between them. Did you find that R and *l* are proportional?

Table-3

	Length of wire *l*	**Electric current I**	**Potential difference V**	**Resistance (V/I)**
3(a)	100 cm			
	80 cm			
	60 cm			
	40 cm			
	20 cm			
3(b)	Cross section area			
	Thin wire (10 cm)			
	Thick wire (20 cm)			
3(c)	Material of wire			
	Nichrome (100 cm)			
	Copper (100 cm)			
	Steel (100 cm)			

(Part 2 – Cross section area of the conducting wire)

- Take two Nichrome wires of 100 cm length but of different thickness.
- One by one tighten the wires tightly, on the mount, and note the current in the wire and the potential difference between the wire ends, and fill in the values in section-3(b) of table-3.
- Establish the relationship between the cross section of the wire and its resistance.
- Remember that the thicker wire has a larger cross section. Would the resistance of the thicker wire be larger than the thinner wire? What did you observe?

(Part 3 – Nature of the wire material)

- Take Nichrome, copper and steel wires of equal length and thickness.
- Mount them in the circuit, one by one, and fill in the values of I and V, in section-3(c) of table-3.
- Which of the above wires has the highest and lowest resistance?

After the activity, we find that, Resistance $R \propto l$ .....(a)

$R \propto 1/A$ .....(b)

From (a) and (b)

$R \propto l/A$ .....(c)

or, $R = \rho\, l/A$ .....(d)

Here ρ (rho), is a relational constant, which is called the 'resistivity' of the conductor. The SI unit of resistivity is Ohm-metre (Ω-metre). It is an inherent property of all materials. Generally, metals and their alloys have low resistivity, which ranges between $10^{-8}\,\Omega$ to $10^{-6}\,\Omega$, and thus they are good conductors of electricity. The resistivity of metals increases with temperature and decreases with lower temperature. But the high resistivity of some special metal alloys like Nichrome, manganin etc., does not depend on temperature. This is the reason that these alloys are used in appliances and heating devices like electric irons, toasters, ovens etc. Tungsten metal is used to make the filament of electric bulbs. Copper and aluminium are used in the manufacture of wires and cables for transmission of electricity due to their extremely low resistivity.

The electrical resistivity of any material does not depend on its length or cross section area but is solely dependent on the nature of the material.

If we take a wire of unit length and unit cross section then,

Resistance, $R = \rho\, l/A$, as $l = 1$ and $A = 1$

Thus, $R = \rho$

Therefore, the resistivity of a wire of unit length and unit cross section is equal to its resistance.

The electrical resistivity of any material is inversely proportional to its conductivity. Materials with low resistivity are good conductors of electricity while those with high resistivity are non conductors.

No material is purely a conductor or a non-conductor. Materials can be depicted from high to low conductivity as shown in figure-6.

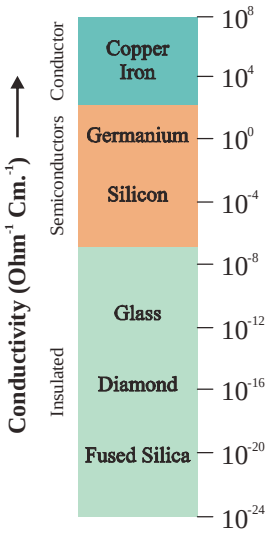


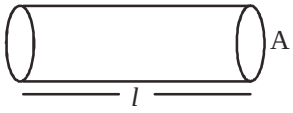
Figure- 6: Increasing conductance of materials

Example-1: If a wire with 8 Ω resistance is melted and a new wire made with double the cross section, what would be the resistance of the new wire?

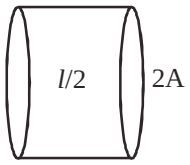
Solution: Given R = 8 Ω, when the thickness of the wire is doubled, as the volume of metal is fixed, the length of the wire decreases by half. We know that –

$$\text{Resistance} = \text{resistivity} \times \frac{\text{length of wire}}{\text{Cross section}}$$

$$R = p\frac{l}{A}, \qquad R_{New} = \rho\frac{\text{Half length}}{\text{Double cross section}} = \rho\frac{l/2}{2A}$$



$$\frac{R_{new}}{R} = \left[p\frac{l/2}{2A}\right] / \left[p\frac{l}{A}\right] = \frac{1}{4}$$



Therefore, $R_{new} = \frac{R}{4} = \frac{8\Omega}{4} = 2\Omega$

The resistance of new wire is 2 Ω.

6.5 Combination of Resistances

As we know, a complete circuit made of many components is required for the flow of current. The circuit components consist of resistance, bulb, cell, connecting wires etc. Whenever a circuit contains more than one resistance, the resistances can be connected in the following two ways –

1. Series combination 2. Parallel combination

6.5.1 Series combination

When in an electrical circuit, the second terminal of a resistor is connected to the first terminal of the second resistor, and the second terminal of this resistor is connected to the first terminal of the third resistor, and so on, this type of combination is known as a series combination. We can extend this series as required. Also, we can add other components like cells, switches, ammeter etc., between the resistors combined in series.

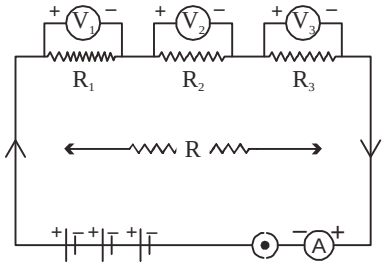


Figure-7: Series combination

In the given diagram three resistors R_1, R_2 and R_3 are combined in a series. If a single resistor replaces all the resistors in such a way that the value of current flowing in the circuit remains the same, then this single resistor is known as equivalent or effectual resistance.

In figure-7, if the current flowing in the circuit is I, then the same amount of current I will flow across each of the resistors. As we know, each point in a circuit has a different potential difference, therefore, each resistor combined in series will have different values of potential difference. If potential difference of resistor R_1 is V_1, R_2 is V_2 and of R_3 is V_3, then as per Ohm's law:

V = IR

Therefore, $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$ .....(i)

The total potential difference in a series is the sum of individual potential difference across each resistance.

$V = V_1 + V_2 + V_3$ .....(ii)

According to equations (i) and (ii),

$IR = IR_1 + IR_2 + IR_3$

$R = R_1 + R_2 + R_3$

Thus, in a series combination the total resistance is equal to the sum of all resistances. Equivalent resistance of the series is therefore equal to R.

The main drawback of a series combination is that if any component of the circuit malfunctions, the flow of current is broken at that point and the whole circuit becomes defective and no component works.

Similarly, ammeter and voltmeter also offer resistance to current flow. The resistance of ammeter is very low and therefore it is connected in series in a circuit so that maximum current flows through it. But the voltmeter is always connected in parallel, because of its higher resistance, so that less current passes through it.

6.5.2 Parallel Combination

In a parallel combination, the first end of all resistors in the circuit, are joined to one point and the last end of the resistors to another point of the circuit. Such type of combination is known as a parallel circuit. We can increase the number of components in the circuit as required. In figure-8, three resistors R_1, R_2 and R_3 have been joined in parallel combination. In the circuit, battery, switch, ammeter and voltmeter have been added. In a parallel circuit, each resistor has the same potential difference, but the amount of current flowing in each resistor is different.

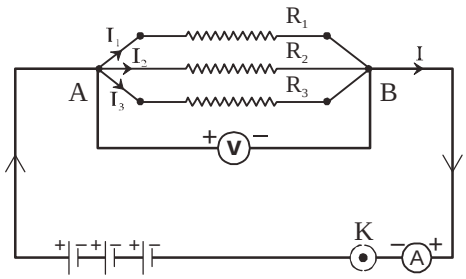


Figure-8: Parallel circuit combination

If in resistors R_1, R_2 and R_3 the amount of current flowing is I_1, I_2 and I_3, respectively, then according to Ohm's law,

$$I_1 = \frac{V}{R_1},\ I_2 = \frac{V}{R_2},\ I_3 = \frac{V}{R_3}$$

If the equivalent resistance of the three resistors is R, then,

$$I = \frac{V}{R}$$

But, $I = I_1 + I_2 + I_3$ (In parallel circuit total current is equal to sum of individual currents in each resistor)

In the above equation,

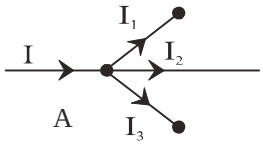
$$\frac{V}{R}=\frac{V}{R_1}+\frac{V}{R_2}+\frac{V}{R_3}$$

Equivalent resistance, $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$

As you have seen, the equivalent resistance in a series combination is greater than the largest resistance in the circuit, but in a parallel combination the equivalent resistance is lower than the smallest resistance in the circuit.

Kirchoff's Rule

According to diagram, a current I passes through point A, and then branches out in three paths I_1, I_2 and I_3.



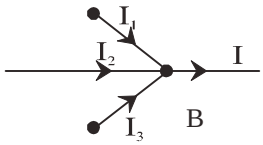
Therefore at point A, $I = I_1 + I_2 + I_3$

$$I - I_1 - I_2 - I_3 = 0$$

At point B, the three branched paths join together and the value of current becomes equal to the value at point A.

At point B, $I_1 + I_2 + I_3 = I$

$$I_1 + I_2 + I_3 - I = 0$$



This is the mathematical representation of Kirchoff's rule. Therefore, according to Kirchoff, the algebraic sum of current flowing in to any point or junction of a circuit is equivalent to the sum of current flowing out of that point or junction. There is no storage of current at any point in the circuit and it is constantly flowing. This rule of Krichoff is in accordance with the law of conservation of charge. This first rule of kirchoff is also known as junction or nodal rule of current.

Discuss :

The old style string of festival lights consisted of all bulbs joined in a series. If even a single bulb fused or malfunctioned, the whole string would stop working. Why? If all appliances in our home are connected in series, what would happen?

Questions :

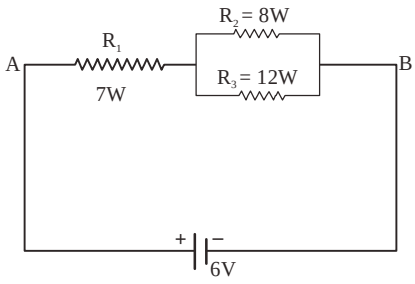
1. If two wires made of the same material, one thick and the other thin, are connected to an equivalent source of current, then, in which wire would the current flow more freely and why?
2. The value of two resistors in a series is in the ratio of 1:4, and a 10 ampere current flows in the circuit. If these two resistors are combined in a parallel circuit, then calculate the ratio of the current flowing through the resistors.

Example-2: Answer the following questions pertaining to the circuit diagram given alongside.

(a) Equivalent resistance of the circuit?

(b) Total current flowing in the circuit?

Solution : (a) We need to calculate the equivalent resistance of the resistances R_2 and R_3 combined in parallel,

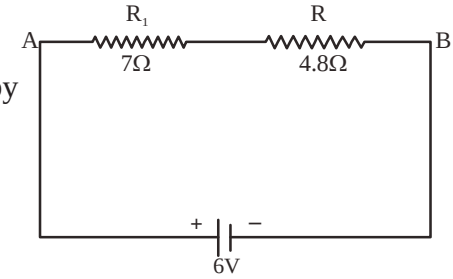


$$\frac{1}{R} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{8} + \frac{1}{12} = \frac{3+2}{24} = \frac{5}{24}$$

$$R = \frac{24}{5} \text{ Ohm} = 4.8 \text{ Ohm}$$

(b) Now, we will calculate the equivalent resistance according to the new circuit diagram given alongside, where the equivalent resistance R, of the parallel resistances R_2 and R_3, has been combined in series with resistance R_1. The equivalent resistance now is,



$R' = R_1 + R$ $R' = 7\Omega + 4.8\ \Omega$ $= 11.8\Omega$

The potential difference created in the circuit by the battery,

V= 6 volts

Equivalent resistance of the circuit = 11.8Ω

As per Ohm's law,

I = 6/11.8

= 0.508 A (ampere)

6.6 Thermal Effect of Electric Current

We use many different electrical appliances in our daily life and are aware that they generate heat when used for a prolonged time. For example, bulbs and motors become hot after some time.

When current flows in any conductor, some electric energy is lost. As per the law of energy conservation, energy cannot be created or destroyed, but can only be converted from one form to another. Therefore, the electric energy last is transformed into heat energy. This effect is known as the thermal effect of electric current. The heat generated in a circuit is due to this thermal effect of current.

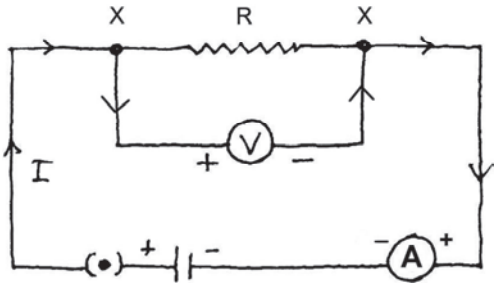


Figure- 9: Joules law of heating

Thermal Rule of Joule: Given that, I current flows for t seconds across a resistance R, and a potential difference V is set up between the two terminals of the resistance,

The charge flowing in the wire in t seconds, = current × time

or, $Q = I \times t$ …….(i)

From the definition of potential difference, the work done to move a charge q, in time t, across the two terminals is,

$W = QV$ and $Q = I \times t$ equation (i)

Therefore, $W = V \times It$ …..(ii)

The work done to move charge q from one end to the other end across the resistance R, manifests itself as the heat generated in the resistance.

Thus, in eq.(ii), if heat H, is substituted for W, $H = VIt$ ….(iii)

Therefore, the heat generated H in the resistance by current I (in time t) is as follows: $H = I^2Rt$ (V=IR, as per Ohms law)

This is known as the Joule's law of heating.

According to this law,

1. The heat H, generated in the resistance, is directly proportional to the square of the current I, flowing across it. Thus $H \propto I^2$
2. The heat generated in the resistance is directly proportional to the resistivity R.

   Thus, $H \propto R$
3. The heat generated in the resistance is directly proportional to the time t, of current flow.

   Thus, $H \propto t$

   $H = I^2Rt$ ….(iv)

6.6.1 Electric Power

The rate at which electric energy is converted into any other form of energy is known as electric power. It is equal to the product of the electric current and potential difference. The rate of loss of energy in any electric circuit is also called electric power. In other words, the rate of work done is known as Power and is denoted by P.

Assume that, in an electric circuit, current I flows across a potential difference V, then the loss of electric energy in t seconds, W = VIt.

Therefore, the loss of energy in 1 second = $\frac{W}{t} = \frac{VIt}{t} = VI$

This is known as Electric Power

$$P = \frac{W}{t} = VI \quad \text{.....(v)}$$

The SI unit of electric power is watt.

From eq.(v), the energy produced in a circuit by a source in time t, is P × t, which equals VIt.

H = VIt .....from (iii)

$H = I^2Rt$ (V = IR)

The value of electricity consumed, H is represented by the unit RWh.

1 kilowatt hour = 1 unit = 1000 watt × 1 hour

1000 watt × 3600 seconds = 36×10^5 watt second

= 3.6×10^6 watt second = 3.6×10^6 Joule

The electric energy consumed is measured by the following equation-

Electricity consumed (in units) = [No. of electric appliances × power of appliance (watt) × Time (hours) × No. of days]/1000

Example-3: In a house, if 4 bulbs of 60 watt burn for 4 hours daily, and 2 fans of 80 watt are used for 10 hours daily, then calculate the electricity consumed in one month and the value of consumed electricity if the rate of electric power is Rs. 2 per unit.

Solution: The electric energy consumed by 4 bulbs of 60 watts used 4 hours per day,in a month

$$= \frac{4\times60\times4\times30}{1000} = 28.8 \text{ units or } 28.8 \text{ Kwh}$$

and energy consumed by the 2 fans

$$\frac{2\times80\times10\times30}{1000} = 48 \text{ kwh}$$

therefore, total electric consumption = 28.8 + 48 = 76.8 units/month

Cost of 1 kilowatt hour = Rs. 2.00

76.8 × 2 = Rs. 153.6 = Cost of consumed electricity

Example-4: If 100 joule heat is being generated by a 4Ω resistor then calculate the potential difference between the two terminals of the resistor.

Solution: H = 100 joule, R = 4Ω , t = 1 second, V = Ω

According to Joule's law of heating

$H = I^2Rt$

$$I^2 = \frac{H}{R\times t}$$

$$I = \sqrt{\frac{H}{Rt}} = \sqrt{\frac{100}{4\times1}} = \sqrt{25} \text{ ampere} = 5 \text{ ampere}$$

Therefore, according to Ohm's law,

V = IR = 5 × 4 = 20 volts

Note: In an electric bulb the filament is made of a thin tungsten metal wire which appears short in length, but actually it is 6 feet long and is made by tightly coiling the wire.

6.7 Domestic Electric Circuit

We receive our electric power in our homes, from the main lines of electric supply. It reaches in our homes by means of a 3 wired electric cable. The three wires of the cable are colour coded. Red is for live current, blue or black for neutral phase and green wire is for earth. In our country, the potential difference between the current (live) wire and neutral wire is 220 volts. The earth wire is of thick copper, and the end of it is joined to a copper plate which is buried deeply underground. It is then joined to the main switch through the electric meter. The live current wire and the neutral wire coming from the electric supply pole, are first attached to a stationery box housing the electric meter. The whole circuit is protected from current overload or short circuit by using fuses combined in a series. If the live current wire and neutral wire get accidentally combined, then the resistance of the circuit becomes null (zero), and heavy current flows in the circuit. This leads to damage of electric appliances in the circuit and can produce sparking and lead to fire. Fuse is a wire made of any metal or metal alloy which has a particular melting point, for example, aluminium, copper, iron, lead etc. If the current flowing in the circuit exceeds a particular value, then the fuse wire heats up and melts. Thus, the circuit is broken and current flow stops. In domestic circuits, usually fuse wires with current ratings of 2A, 3A, 5A or 10A are used. The electric current is transmitted, from the power producing station, in the form of alternating current (A.C.). In this form of current, the direction of the current flow is constantly changing. For domestic use, current of 5A, and for industrial use 15A is supplied and it has a frequency of 50 Hertz.

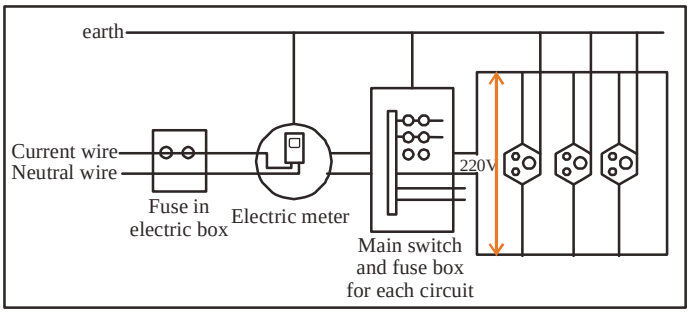


Figure-10: Domestic electric circuit

The current produced by a cell or battery has a fixed direction and its frequency is zero hertz. This is known as Direct Current (D.C.).

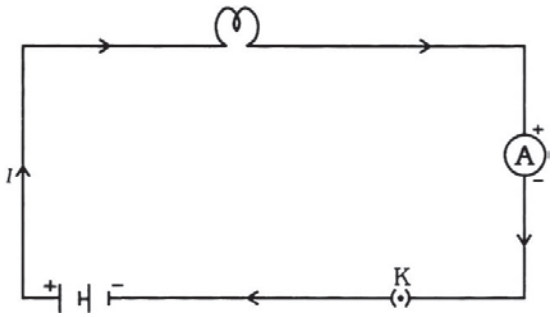
6.8 Precautions while using Electricity

As we have seen, there is a danger of fire in an electric circuit due to overload or short circuit. Therefore, the following precautions should be taken during use:

- Switches and plugs should be fitted tightly.
- Wires having proper covering insulation should be used.
- Damaged or wires in poor condition should be immediately replaced.
- The circuit should be immediately switched off in case of fire or any accident.
- All circuits should be protected by fuse wires of proper material and rating.
- All electric appliances should be properly earthed.
- Never touch any part of the circuit with wet hands.

Solved Numerical Problems

1 In a electric bulb, a current of 0.5 A flows for 10 minutes. Calculate the electric charge flowing in the electric circuit?



Solution: Given that, I=0.5 A, t=10 minutes = 600s

$Q = I \times t$

$= 0.5\ A \times 600\ s$

$= 300\ C$

2. A metallic wire of length 1 meter has a resistance of 26Ω at a temperature of 20°C. If the diameter of the wire is 0.3 mm, then what is the resistivity of the metal composing the wire at the given temperature?

Solution: Given that the wire Resistance, R = 26Ω, dia, = 0.3 mm = 3×10^{-4}m, r = $\frac{3\times10^{-4}}{2}$

= 1.5×10^{-4} and length of wire l = 1 m

Therefore, according to the equation, the resistivity of the wire,

$$\rho = (RA/l) = (R\pi d^2/4l) = \frac{R\pi r^2}{l} = \frac{26\times22\times1.5\times10^{-4}\times1.5\times10^{-4}}{7\times1}$$

By solving the above, we get,

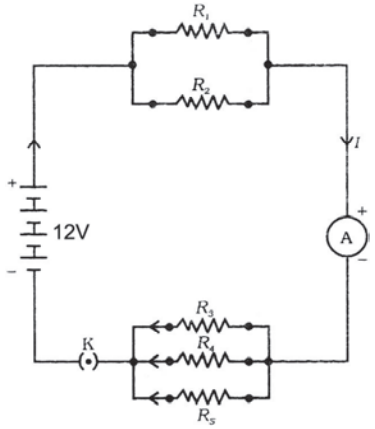
$\rho = 1.84 \times 10^{-6}\ \Omega$ m

Thus, the electric resistivity Ω, of the wire metal at 20°C is $1.84 \times 10^{-6}\ \Omega$ m

3. In the diagram given alongside, $R_1 = 10\Omega$, $R_2 = 40\Omega$, $R_3 = 30\Omega$, $R_4 = 20\Omega$ and $R_5 = 60\Omega$. This configuration of resistors are connected to a 12 V battery. Find out

(a) Total resistance in the circuit, (b) The total current in the circuit.

Solution: Let us substitute the combinations of resistors R_1, R_2 and resistors R_3, R_4, R_5 with equivalent resitances R' and R" respectively. Then, by using the equation, we get,



1/R' = 1/10 + 1/40 = 5/40 i.e., R' = 8Ω

Similarly, 1/R" = 1/30 + 1/20 + 1/60 = 6/60

i.e., R" = 10Ω

Therefore, total resistance R = R' + R" = 8Ω + 10Ω = 18Ω

To calculate the total current using Ohms law, we get,

I = V/R = 12V/18Ω = 0.67 A

4. In an electric iron, at the highest thermostat setting, electric power is consumed at 840 watt and at the lowest setting at 360 W. If the source voltage is 220 V, then what is the resistance in the two situations?

Solution: We know that power consumed,

$P = VI$

So, $I = P/V$

(a) When the rate of heating is highest, then $I = 840\ W/220\ V = 3.82\ A$

Therefore, the resistance of the iron is, $R = V/I = 220\ V/3.82\ A = 57.60\ \Omega$

(b) When the rate of heating is lowest, then, $I = 360\ W/220\ V = 1.64\ A$

Therefore, the resistance of the electric iron is, $R = V/I = 220\ V/1.64\ A = 134.15\ \Omega$

Keywords

Electric current, electric potential, electrical potential difference, ammeter, voltmeter, resistance, resistivity, electric non conductor, equivalent resistance, electric power, direct current, alternating current, fuse, overload, short circuit, semi conductor, variable resistance



What we have Learnt

- For electric current to flow in any conductor, a closed circuit is necessary.
- The quantity of electric current in a conductor is the amount of electric charge that flows in 1 second through any point of the conductor.
- To measure the electric current in a circuit we use the ammeter, which is combined in a series in the circuit.
- A potential difference must be present for electric current to flow in any electric circuit.
- To measure the potential difference in any electrical circuit, we use an instrument called the Voltmeter, which is attached in parallel combination in the circuit.
- According to Ohm's law, in any electric circuit, at manual temperature, the potential difference and electric current flowing in two terminals of a wire are directly proportional i.e. $V \propto I$ or $V=IR$.
- Any resistance offered by a conductor, shows that it has resistance to the flow of current through it. This is called electric resistance and denoted by R.
- ρ (rho) is a relational constant, which is termed as the resistivity of the material of the conductor. It solely depends on the material composition and temperature.
- The equivalent resistance of resistors attached in series combination, is given by $R = R_1 + R_2 + R_3$..... and so on. The same amount of current flows in each resistor arranged in a series combination, but the potential difference between any two points of the circuit varies.

- Resistors combined in parallel have an equivalent resistance $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$ and in each resistor different quantity of current flows but the potential difference between the two terminals of each resistor is the same.
- The rate at which electric energy is converted into another form of energy is called electric Power. Electric power is equivalent to the product of electric current and potential difference, i.e. P = V×I

Exercises

1. Choose the correct option :

   (i) Which one of the options given below is a good conductor of electricity–

   (a) Wood piece (b) Sheet of paper (c) Copper wire (d) glass rod

   (ii) The instrument used to measure current in a circuit is–

   (a) voltmeter (b) ammeter (c) odometer (d) none of these

   (iii) The relation between V and I for a conducting wire is–

   (a) a variable ratio (b) a fixed ratio

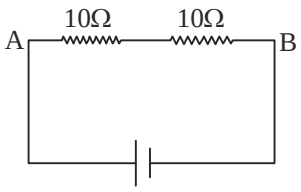
   (c) sometimes variable sometime fixed (d) none of these

   (iv) The resistance of an electrical instrument is 2.2 ohm and potential difference is 220 volts. Find out the value of current on connecting the instrument to a electric source–

   (a) 5 A (b) 20 A (c) 25 A (d) 10 A

   (v) The equivalent resistance of the following diagram would be–

   (a) 25 Ω (b) 20 Ω (c) 10 Ω (d) 35 Ω



2. Fill in the blanks–

   (i) The potential difference of resistors connected in a series would be .....

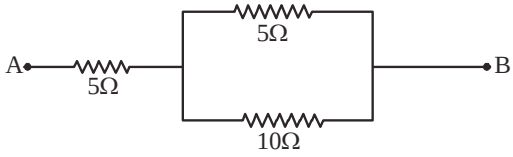
   (ii) In resistors connected in a parallel combination, the current flowing across each resistor would be .....

   (iii) We show the consumption of electricity in Kwh or in .....

   (iv) The filament used in electric bulbs is made of ..... metal.

   (v) The ..... of a conductor increases with temperature.

3. Find out the equivalent resistance of the circuit given along side– **(8-33 Ω)**



4. What is a fuse and of what material is it made?

5. Explain the Ohm's law in your own words and establish the relation between V & I by drawing a graph.
6. If an electric bulb consumes 2400 J energy in a minute, calculate the power of the bulb? **(40 w)**
7. An electric heater has 3 kw, 220 v, stamped on it. Find values of the following–

   (a) Electric current **(13.6 A)** (b) Resistance of the heater **(16.12 ohm)**

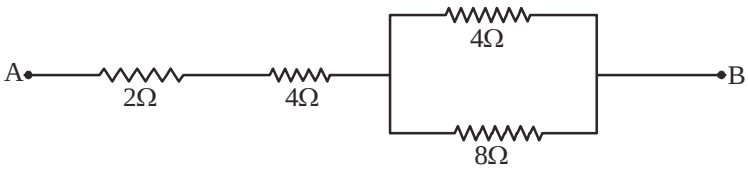
   (c) Cost of electricity if unit cost is Rs. 1.50/kwh and heater is used for 10 hours **(Rs. 45)**
8. Answer the following questions-

   (i) Why are domestic circuits not arranged in a series combination?

   (ii) Why is only tungsten metal used to make filaments of bulbs?
9. If a current of 0.5 A flows for 20 minutes in a filament of a bulb, then what would be the quantity of electrical charge flowing in the circuit?
10. In a house, 4 electric bulbs of 40 watt burn for 5 hours, 2 bulbs of 60 watts for 6 hours and 3 fans of 80 watts run for 6 hours, daily. What would be the cost of electricity, for one month of that home, if the rate is 50 paise per unit? **(Rs 44.4)**
11. Discuss the following questions -

    (i) What would be the effect on the circuit if one bulb out of three bulbs arranged in a series combination becomes fused?

    (ii) What would be the effect if one bulb out of three arranged in parallel combination malfunctions?
12. Find out the equivalent resistance between A & B in the circuit diagram given below? **(8.66Ω)**



13. If the potential difference generated by a battery across a circuit of two bulbs, combined in series, is 6 volts, and the potential difference at the first bulb is 2 volts, find out the potential difference at the second bulb? **(4 volts)**
14. If two conducting wires of the same length and cross section are combined first in a series and then in a parallel combination, what would be the ratio between the quantity of heat generated in the two series and parallel circuits? **(4:1)**
15. A 9 volt battery is arranged in series combination with resistors of 2Ω, 3Ω, 4Ω, 5Ω and 12Ω, what would be the current flowing across the 2Ω resistor?
16. List the various appliances working on the thermal effect produced by the electric current, which are used in our homes, and discuss their working?
17. On a bulb, 200 V - 100 W is printed. What would be the resistance of the bulb? If 5 of these bulbs burn for 4 hours, what would be the amount of electricity consumed and the cost of electricity, if the rate is 50 paise per unit? **(400Ω, 2Kwh, Rs.1)**
18. How would you combine 3 resistances of 2Ω, 3Ω, and 5Ω, so that the equivalent resistance is 2.5Ω. Draw a diagram to show the circuit.